EREN HASAN KASAP

Mathematical Engineer Student

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EDUCATION

Yıldız Technical University

Mathematical Engineering (100% English)

2024-2028(Expected)

Izmir University of Economics

Computer Engineering
(100% Scholarship)(100% English)

2023-2024

TECHNICAL SKILLS

Programming Languages

TypeScript, JavaScript, Python, C, HTML5, CSS3

Technologies & Frameworks

Next.js, React Native, Vue.js, Tailwind CSS, NativeWind, Expo, PyTorch

Systems & Tools

Docker, Linux/UNIX, Github/Git, npm

Concepts

Mathematical Modeling, Optimization Algorithms (Gradient Descent), System Programming, Data Analysis (Pandas, NumPy)

LANGUAGES

- Turkish : Native
- English : Advanced (C1)

ABOUT ME & OBJECTIVE

Mathematical Engineering student focused on engineering real-life implementations. Adept at utilizing mathematical models, algorithms, and software architecture not just as end goals, but as primary tools to optimize processes and create data-driven, practical solutions for complex engineering problems.

PROJECTS

Personal Portfolio & Technical Blog | Completed | Next.js, TypeScript, Tailwind CSS | January 2026

- Architected and deployed a production-grade portfolio website to demonstrate "real-life production" capabilities over theoretical knowledge.
- Implemented a modern tech stack to overcome standard CSS limitations, focusing on efficient feature integration and mastering Next.js architecture.

Algorithmic Mobile Game Architecture | Processing | GoLang, Docker, React Native | January 2026 – Present

- Collaborating on the development of a cross-platform mobile game, prioritizing scalable backend integration and application security.
- Containerizing backend environments using Docker and implementing foundational system programming concepts for performance optimization.

Fast Gradient Descent Optimization Research | Processing | Python, PyTorch, Mathematics | October 2025 - Present

- Developed a TÜBİTAK 2209A project proposal focusing on a hybrid approach to Fast Gradient Descent optimization algorithms.
- Researched and modeled mathematical frameworks to accelerate convergence rates in quantitative analysis and machine learning contexts.

React Native Mobile Application | Failed | React Native, Expo, TypeScript, NativeWind | Collaborative Project | August 2025

- Developed and implemented UI/UX for a cross-platform mobile application using React Native and TypeScript.
- Integrated RESTful APIs to connect with backend services for data fetching and user interactions.
- Collaborated with backend developer to ensure seamless API integration.
- Project development was discontinued due to coordination challenges with partners and a significant decline in target market demand.

VOLUNTEER WORKS

Frontend Developer | ESTIEM (European Students of Industrial Engineering and Management) | September 2024 - Present

- Engineered structural modifications and restructured data flow architecture to improve overall system performance and maintainability.
- Spearheaded the transition from resolving minor UI/UX issues to leading comprehensive problem identification and replacing legacy testing technologies.

Vice President of Corporate Relations | ESTIEM (European Students of Industrial Engineering and Management) Local Group of Izmir-Economy | May 2024 - September 2024

- Orchestrated two major corporate events, successfully securing sponsorships and executing strict financial administration and budget calculations.
- Managed committee operations, bridging communication between student teams and corporate partners.